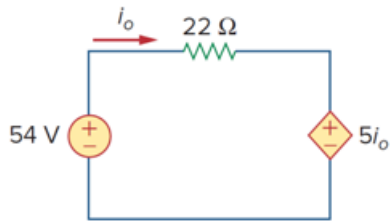


Problem

Determine i_o in the circuit of Fig. 2.84.

Figure 2.84:



Step-by-step solution

Step 1 of 2

The circuit is shown in Figure 1.

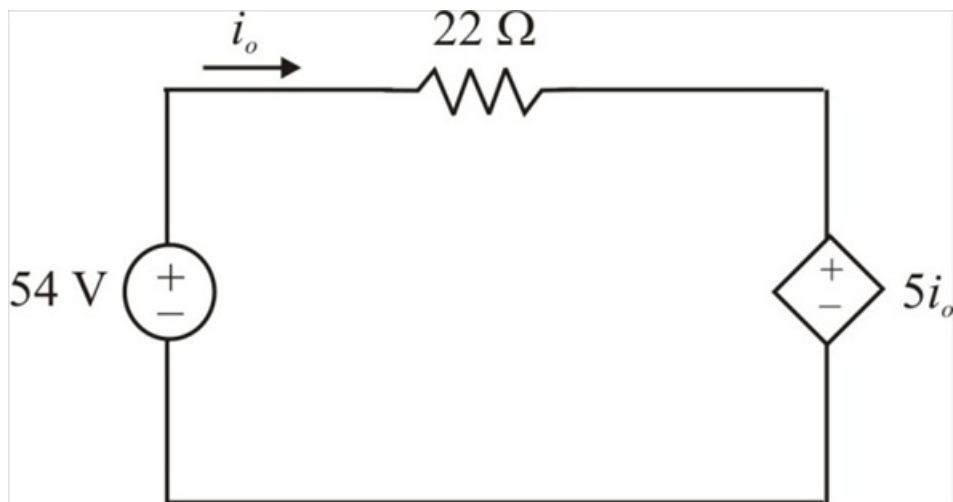


Figure 1

Step 2 of 2

Apply Kirchhoff's voltage law, the algebraic sum of all voltages around a closed path (or loop) is zero. Calculate the current i_o .

$$-54 + 22i_o + 5i_o = 0$$

$$27i_o = 54$$

$$i_o = 2 \text{ A}$$

Therefore, the current i_o is 2 A.